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A three-stage theory of international expansion: the link between multinationality and performance in the service sector

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Abstract

It is generally assumed that the performance of a firm improves with greater multinationality. Yet recent empirical studies have shown both a U-shaped relationship (which suggests an initially negative effect of international expansion on performance, before the positive returns of international expansion are realized) and an inverted-U-shaped relationship (which suggests that international expansion beyond an optimal level is again detrimental to performance, and results in a negative slope). This paper proposes a new unified three-stage theory of international expansion that incorporates both concepts in a sigmoid hypothesis. It then tests this on data from 11 service industries, highlighting the difference between knowledge-based and capital-intensive service sectors.

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Introduction

Is international expansion good for a firm?

The foundation of international business studies rests on the assumption that increased multinationality is good for a firm's performance. Vernon (1971) asserted a positive relationship between performance indicators such as return on investment (ROI) or return on sales (ROS) and the extent of multinationality of the firm. International expansion allows the firm to capture economies of scale, or geographic scope (Kogut, 1985). Dunning (1993) averred that less saturated foreign markets provide companies with the means to maintain and expand distribution and gain overall market share by exploiting their current stock of assets – that companies with valuable transaction-based ownership advantages (O_i) can reap internalization benefits, circumvent market failure, and avoid trade barriers, moral hazards, and broken contracts.

Advantages of international expansion accrue by:

- (1) Spreading common and central overheads over more and more nations: this is especially critical in R&D-intensive industries that require amortization of R&D from more than a few markets (Kobrin, 1991; Tallman and Li, 1996).

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- (2) Greater learning or international experience (Kobrin, 1991).
- (3) Access to cheaper and idiosyncratic resources in foreign countries: these could include cheaper labor, better technology, or any country-specific resource (Porter, 1990; Jung, 1991).
- (4) The abilities of the multinational enterprise (MNE) for global scanning of rivals, markets, and other profit opportunities.
- (5) Better cross-subsidization, price discrimination, and arbitrage potential with larger geographic scope.

In general, the greater the number of countries the MNE serves, the better its *appropriability regime* (Teece, 1986).

Thus core international business theory argues for a positive relationship between the performance of the firm (measured by a variety of indicators such as ROI, ROS, ROA, ROE, growth, and Tobin's Q; see Table 1) and its degree of multinationality.

But can international expansion produce a diminution in performance?

For the most part, international expansion is positive for a firm's performance. Yet recent

empirical studies have shown both a U-shaped relationship (which suggests an initially *negative* effect of international expansion on performance, before the positive returns of international expansion are realized) and an *inverted* U-shaped relationship (which suggests that international expansion beyond an optimal level is detrimental to performance, and again results in a *negative* slope if firms are excessively internationalized).

The overall shape of the relationship between the performance of MNEs and their degree of multinationality is a crucial question for academics, as well as for company strategists who are concerned about (1) initial internationalization barriers and (2) for mature multinationals, the 'optimum' number of countries that the firm serves, beyond which further expansion may be sub-optimal. The existing literature, summarized in Table 1, shows linear, U-shaped, and inverted-U-shaped relationships.

This paper's contribution is to:

- (1) reconcile these seemingly contrary findings by proposing a new three-stage theory of international expansion;
- (2) test this theory using data from 11 service industries in 12 nations; and

Table 1 Previous literature on the link between performance and degree of multinationality

	Author(s) and year	Performance indicators
Linear	Han <i>et al.</i> (1998) (+)	ROE, asset turnover, profit margin
	Jung (1991) (+)	(After-tax net income)/(Total assets)
	Vernon (1971) (+)	ROI, ROS
	Kim and Lyn (1987) (+)	Excess market value; Tobin's Q
	Errunza and Senbet (1981) (+)	Excess return
	Grant (1987) (+)	ROA, ROE, ROS
	Grant <i>et al.</i> (1988) (+)	ROA, ROE, ROS
	Brewer (1981) (–)	Stock return
	Siddharthan and Lall (1982) (–)	Sales growth
	Michel and Shaked (1986) (–)	Risk-adjusted return
	Collins (1990) (–)	Total risk, Debt to equity ratio, Beta
	Buckley <i>et al.</i> (1977, 1984) (0)	ROA
	Kumar (1984) (0)	ROA, ROS
	Morck and Yeung (1991) (0)	Market value
	Qian (1997)	ROE
U-shaped	Ruigrok and Wagner (2002)	ROA
Inverted-U-shaped	Daniels and Bracker (1989)	ROA, ROS
	Geringer <i>et al.</i> (1989)	ROA, ROS
	Sullivan (1994a, b)	ROA, ROS
	Ramaswamy (1995)	ROA, ROS, ROVA
	Al-Obaidan and Scully (1995)	Frontier production function, Variance in technical efficiency
	Gomes and Ramaswamy (1999)	Cost of sales/total sales, ROA
	Hitt <i>et al.</i> (1994)	ROA, ROS

Findings: +, positive relationship; –, negative relationship; 0, no relationship.

- (3) shed new light on the growth of service firms by highlighting differences between knowledge-based and capital-intensive service sectors.

Prior studies on the multinationality-performance nexus

The studies in Table 1 that show a positive linear fit confirm the bedrock proposal of international business studies – that a firm's performance improves with greater international expansion.

A negative linear result suggests a decline in performance, perhaps with initial internationalization.

A U-shaped relationship suggests that *early internationalizers* see an initial decline in performance up to a point, after which further expansion brings an upward performance in returns from increasing geographic scale.

An inverted-U-shaped relationship suggests that geographic expansion leads to enhanced corporate performance (positive slope) up to a second 'threshold', beyond which the organizational costs and complexity associated with managing widely scattered operations begin to outweigh the advantages of even further international expansion (that is, a negative slope sets in again).

Why these seemingly contradictory findings? In some studies it may simply be that a quadratic term was not introduced into the equation, and that a linear fit was statistically significant. In other cases the data may be capturing only *part* of an overall sigmoid (S-shaped) function, as proposed in Figure 1. Depending on which part of Figure 1 we examined, we can find linear (positive

and negative), U-shaped, and inverted-U-shaped segments.

A new three-stage theory of international expansion

Stage 1. Negative slope: costs and barriers to initial international expansion

Implicit in the Uppsala school's 'internationalization theory' hypothesis (Johanson and Vahlne, 1977), that a firm initially seeks only familiar markets, is the little-explored notion of the *liability of foreignness* – the additional burden, or costs, that the internationally expanding firm must initially endure. Zaheer and Mosakowski (1997) couch this in terms of the initial costs of a foreign firm establishing its legitimacy abroad. Johanson and Vahlne (1977) acknowledge the costs of acquiring foreign market knowledge, and virtually say that a company's foreign expansion will not occur before the costs of acquiring knowledge about the foreign market are first incurred. Early internationalizers have large learning costs because of unfamiliarity with foreign markets, cultures and environments. The role of knowledge acquisition costs, relating to foreign markets, is treated more explicitly in a follow-up work by Johanson and Vahlne (1990). For early internationalizers, as the initial scale of global operations is small, the up-front costs of creating an international operation are not yet recouped from the relatively few nations in which the firm operates (Hitt *et al*, 1997; Gongming, 1998). Going international entails a large minimum administrative overhead burden, which, if spread over just a handful of country markets, results in a high burden of overhead per nation burden. In Stage 1,

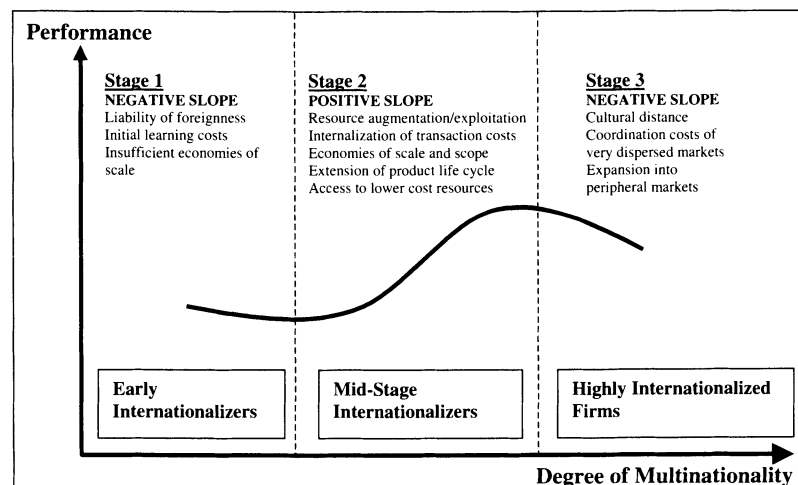


Figure 1 A three-stage sigmoid (S-shaped) hypothesis.



therefore, we expect a negative slope for the link between performance and multinationality, as shown in Figure 1.

However, the costs and barriers of early internationalization are not assumed to be onerous – otherwise few firms would venture abroad. Hence Stage 1, in Figure 1, is hypothesized to have a relatively shallow negative slope, and to be of shorter extent than Stage 2.

Stage 2. Positive slope: benefits of international expansion are now realized

In mid-stage international expansion, further geographical scale makes possible efficiencies that improve performance indicators such as ROS, or overheads per nation. The fixed costs and overhead burden of headquarters operations and large R&D outlays can be increasingly spread over more nations (Kogut, 1985; Porter, 1985). The incremental benefits of further international expansion are now greater than the incremental costs of further Stage 2 expansion.

For both market-exploiting and resource-seeking MNEs, the greater the number of countries, the more the firm can engage in price discrimination, strategic cross-subsidization, and arbitrage (Contractor, 2002). The multinational firm exploits its ability to arbitrage national differences, and the larger its global scope, the greater is this ability (Rugman, 1981).

Resource-seeking companies are better able to access low-cost inputs, including labor, and to tap into knowledge clusters (Daniels and Bracker, 1989; Annavarjula and Beldona, 2000). The transfer of specialized learning from certain nations – or, in general, the ability to cross-fertilize knowledge across subsidiaries – increases with increased multinationality. Similarly, market-seeking firms are better able to scan for market opportunities. Other benefits of Stage 2 international expansion are the ability of some companies to exercise global market power (Grant, 1987) and to extend the product cycle (Vernon, 1966).

Hence the commonly accepted hypothesis that, *ceteris paribus*, multinationality is positively associated with performance, in Stage 2.

Stage 3. Negative slope: international expansion beyond an optimal threshold

But are the benefits of further international expansion *indefinitely* positive? The many *inverted-U*-shaped results shown in Table 1 suggest strongly that some companies in a sector, or a sample, *over-*

expand internationally beyond a desirable optimum level. For such firms, the incremental costs of further expansion into peripheral nations are greater than the incremental benefits, and hence are detrimental to overall global performance. The performance–multinationality slope again becomes negative, as seen in Figure 1. Why? First, beyond a certain point, having expanded into the most lucrative markets, the firm is then left with minor or peripheral countries with a lower profit potential. Second, beyond an optimum number of nations, the growth of coordination and governance costs may exceed the benefits of further expansion, because of the complexity of global operations (Galbraith and Kazanjian, 1986). This is especially true as the number of different cultural environments that the firm has to deal with increases transaction and governance costs (Gomes and Ramaswamy, 1999). Grant (1987) suggests that ‘limits to the capacity of managers to cope successfully with greater complexity’ may inhibit the indefinite (that is, monotonic) realization of the net positive value of international expansion. Siddharthan and Lall (1982) also suggest that, while firms can achieve economies of scale as a result of FDI, excessive multinationality may lead to increased managerial constraints due to legal barriers, and to physical, cultural, and linguistic distance. Hitt *et al.* (1990) describe this in terms of loss of strategic control in some overextended companies because of high information costs. In short, taking an ‘incremental benefit = incremental cost’ approach, beyond some level of multinationality, ‘the coordination required (for multiple transactions among many geographically diverse units) may cost more than the benefits derived from sharing resources and exploiting market opportunities’ (Hitt *et al.*, 1997). That is to say, Stage 3 hypothesizes, once again, a negative relationship for the link between performance and multinationality.

However, *a priori*, we do not expect that many firms will be thus over-internationalized: hence a subordinate hypothesis, that Stage 3 will be shorter than Stage 2 (as drawn in Figure 1).

But why would even a few firms overextend themselves into the sub-optimal stage 3?

The many inverted-U-shaped results from past studies indicated in Table 1 show that at least a few firms stray into a sub-optimal zone for performance. Why?

First, it is difficult for a firm to assess when it is over-internationalized (unless perhaps it undertakes studies such as this one – this is a clear managerial prescription of this paper). But an academic cross-sectional exercise to reveal a firm's position in relation to its competitors, or to a sector, is not something that companies usually undertake. Hence many firms simply do not know when they are overextended. International expansion in firms is rarely the result of continuously monitored, paced growth, but is based on discrete choices.

Second, some firms may *deliberately* over-internationalize, for long-term strategy reasons such as market share, or for tapping into knowledge clusters, even though this is detrimental to medium-run performance as measured by ROA, ROS or ROE in the cross-sectional studies shown in Table 1. The long-run strategic value of such deliberate over-internationalization may arguably be reflected in improved stock market value, but only if we believe that stock prices are based on rationality and full information. Moreover, in the studies listed in Table 1, only three were based on stock market value; the rest used indicators such as ROA or ROS.

A new international expansion theory: an overall sigmoid (cubic) hypothesis

Past studies, as shown in Table 1, have used diverse data sets from a variety of sectors. Those that have found a linear positive slope either have not specified a quadratic term, or may simply be capturing Stage 2 (mid-stage internationalizers) of Figure 1 in their data and results. Studies that show a negatively sloped linear function (see the Appendix table at the end of this paper for an example) may be capturing only Stage 1 (early internationalizers) in Figure 1. Similarly, studies that have found a U-shaped curve may reflect Stages 1 and 2 *combined*. Finally, several studies shown in Table 1 had an inverted-U-shaped result (which captures Stages 2 and 3 *combined*) because they merely specified a quadratic function.

This paper will specify a cubic regression term to see whether *all three stages* exist, as empirical fact. Nevertheless, there may be subsectors that are not fully internationalized: if the degree of multinationality in a certain subsector is not complete (most companies are in Stages 1 and 2, and few firms have reached their optimal degree of internationalization, or have crossed into Stage 3) then the shape of the statistically fitted curve, despite a

cubic or sigmoid regression specification, would only be U-shaped. By the same token, in samples that contain mainly mature or highly internationalized companies (Stages 2 and 3) the statistically fitted curve may be an inverted-U-shaped curve, despite a cubic term.

But the meta-hypothesis is that all three stages exist, showing: (1) a short negative slope, then (2) a positive slope, and then (3) a short negative slope again, as illustrated in Figure 1.

The service sector

All the past studies listed in Table 1 are in manufacturing; but services are more important. By 1990 the share of services in GDP had reached 61 and 45% for developed and developing countries, respectively (World Bank, 1992). The phenomenal growth of service multinationals in the past decade in accountancy, advertising, banking, consultancy, hotels, insurance, legal, telecommunications, and other service sectors is not yet reflected in academic studies.

Although services are more important than manufacturing, there is little research on the growth and internationalization of service firms. Moreover, services are fundamentally different from manufacture, in terms of relative intangibility, perishability, simultaneity of production and consumption, and customization (Boddewyn *et al.*, 1986). Not that all service sectors are homogeneous. As we shall see later in the paper, there are substantial differences among different types of services in terms of capital intensity and knowledge intensity, which produce different results in terms of the link between performance and multinationality.

The data

The data were taken from *Directory of the World's Largest Service Companies*, published by the United Nations Center on Transnational Corporation, and Moody's Investors Service in 1990. (The word 'largest' in the title may be misleading in the context of this paper. The service sector is not, and was not, as internationalized as manufacturing, and so even the largest service firms may have a relatively lower degree of multinationality. Moreover, some service firms are just beginning their internationalization process. However, *a priori*, we expect all three stages of the sigmoid hypothesis to be represented in the sample.) The *Directory* lists over 200 service firms covering 14 service sectors, and contains company demographics

**Table 2** Samples from *Directory of Service Companies* published by United Nations Centre on Transnational Corporations

	Number of companies in the directory	Percentage of U.S.-based companies	Number of companies involved in study	Percentage of U.S.-based companies
1. Accounting	15	60	0 ^a	0
2. Advertising	15	53	10	50
3. Air transport	15	47	11	45
4. Construction	16	38	12	17
5. Hotel	14	71	7	57
6. Legal services	17	47	0 ^a	0
7. Market research	9	67	6	50
8. Publishing	16	56	16	56
9. Rail transport	13	38	0 ^a	0
10. Restaurant and fast food chains	15	100	3	100
11. Retail trade	15	60	6	50
12. Securities and diversified financial services	15	47	11	55
13. Shipping and trucking	14	43	10	30
14. Wholesale trade	15	13	11	0
Total	204	52	103	42

^aNo complete data on international operations.

Table 3 Subsamples involved in this study: knowledge based vs capital based*Knowledge-based service sectors*

Accounting	0 ^a
Advertising	10
Legal services	0 ^a
Market research	6
Publishing	16
Securities and diversified financial services	11
Total subsectors used in regression analysis	4
Total companies used in regression analysis	43
Total observations	254

Capital-based service sectors

Air transport	11
Construction	12
Hotels	7
Rail transport	0 ^a
Restaurant and fast food chains	3
Retail trade	6
Wholesale trade	11
Shipping and trucking	10
Total subsectors used in regression analysis	7
Total companies used in regression analysis	60
Total observations	364

^aSubsector eliminated because of lack of information on international operations.

and financial data for the period 1983–1988. For each service sector, a summary covers its market size and structure. Of the companies in this

directory, 52% were headquartered in the United States. In half of the 14 sectors, more than 50% were US-based – ranging from 100% in the restaurant and fast food sector to less than 13% in wholesale trade, as seen in Tables 2 and 3. Hitt *et al.* (1994) suggest differences in a MNE's abilities, '...depending on their home country or area or cultural circumstance...' To test this, a dummy variable will be introduced for the home country.

Since complete data were lacking in some sectors such as accounting, legal services, and rail transportation, these had to be eliminated from the study, and the final sample consisted of 11 service sectors. Not all companies disclosed complete information on their foreign operations, and so they were removed from the final sample. The final number of firms therefore amounted to 103 out of 204 listed. Of the total sample of companies, 42% were headquartered in the US.

Why one should distinguish between service subsectors?

All subsectors are not the same, either in manufacturing or in services. In their defining work on services, Boddewyn *et al.* (1986) recognized that, within the rubric of 'services', there can be substantial differences in attributes such as intangibility, perishability, and simultaneity. Recent work examining the relationship between performance and multinationality has introduced 'moderating' variables that distinguish between types of firm.



Kotabe *et al.* (2002) show that the ability to extract performance benefits, from the same level of multinationality, '...will vary... based on their capability to maximize the gains of multinationality while minimizing the relevant costs of expansion.' They distinguish between their sample firms based on R&D and marketing intensity. By contrast, the present study distinguishes between capital-intensive service sectors and knowledge-based service sectors, which are more driven by intangible assets and have a much lower fixed capital cost burden.

The 11 service sectors in Table 2 are divided into two groups. *Knowledge-based service sectors* include advertising, marketing research, publishing, securities, and diversified financial services; *capital-intensive service sectors* include air transportation, construction, hotel, restaurant and fast food chains, wholesale trade, retail trade, and shipping and trucking. International expansion strategy also varies across the two subsamples. In knowledge-based service sectors, from the 1980s on, international expansion tended to be driven by a 'follow-the-client' strategy, as advertising, financial services and market research companies followed their multinational clients abroad. We hypothesize that barriers to internationalization (in Stage 1 of Figure 1) will be less onerous for knowledge-based sectors than it will be for opposed to capital-intensive sectors. Hence knowledge-intensive service sectors will be able to reach the Stage 2 (positive slope) advantages of internationalization more quickly than capital-intensive sectors will. In a cross-sectional analysis such as this, this would be inferred from the relative extent, or length, of the negative slope of Stage 1 of the sigmoid curves for knowledge-based and capital-intensive subsamples.

Methodology

The methodology involves pooled cross-section/time-series analysis in a cross-sectionally heteroskedastic, time-wise autoregressive model. This enables researchers to examine variations among cross-sectional units simultaneously with variations within individual units over time (Sayrs, 1989; Hsiao, 1995). It assumes that regression parameters do not change over time (temporal stability) and do not differ between various cross-sectional units (cross-sectional stability). This technique enhances the reliability of the coefficient estimates.

Even assuming that explanatory variables are nonstochastic, a problem may exist when combining the two sets of data, such as nonconstant error variance arising from between-subject differences (Neter *et al.*, 1990). Nerlove (1971) pointed out that autocorrelated errors are frequently encountered, when dealing with time-series analysis, because of correlations among variables over time.

To minimize autocorrelation and heteroskedasticity problems, the Kmenta (1986) autoregressive-heteroskedastic model will be used, as seen in Boeker and Goodstein (1991), Hill and Phan (1991), and Gomes and Ramaswamy (1999). The assumption of a generalized linear regression model in a two-stage generalized least squares regression allows for nonconstant variance and corrects for autocorrelation. Thus this study need not make the restrictive assumption of equal variance among individual service company data.

Models and operationalization: building on previous studies

Grant (1987) identified two types of study on the link between performance and multinationality. The first type investigated whether or not multinational corporations outperformed their domestic rivals, an issue treated by Vernon (1971), Horst (1972), Shaked (1986), and Collins (1990). The second type compared only international companies based on some index such as the extent or degree of the firm's internationalization.

The degree of multinationality has been measured in terms of ratios of foreign to total sales (Grant *et al.*, 1988; Geringer *et al.*, 1989); or of foreign to total assets (Ramaswamy, 1993); or of employees in foreign locations to total employees (Kim *et al.*, 1989). Such financial measures tend to capture the 'depth' of internationalization.

Other studies index the scope or 'breadth' of internationalization by examining the number or geographical dispersion of operations across countries (Kogut, 1985). Ramaswamy (1993) measured multinationality in terms of the number of overseas plants, and found a significant positive relationship to performance. Shaked (1986) defined multinational corporations as having 20% of sales outside the home country and direct investment in at least six countries.

The use of a unidimensional measure of internationalization listed in the above studies has been criticized, and some have called for a multidimensional construct (Sullivan, 1994a, b). However, little

empirical support was found for a multidimensional measure in Ramaswamy *et al.* (1996).

As seen from the above, there is no standard approach for measuring the degree of multinationality. For this study, we constructed a composite index for the degree of internationalization and used a sigmoid, or cubic regression model,

$$\begin{aligned} \text{PERF}_{it} = & \beta_0 + \beta_1 \text{SIZE}_{it} + \beta_2 \text{DOI}_{it} + \beta_3 (\text{DOI}_{it})^2 \\ & + \beta_4 \times (\text{DOI}_{it})^3 + \sum \beta_{4+m} I_m \\ & + \beta_{4+\max(m)+1} C_1 + \varepsilon_{it} \end{aligned}$$

with first-, second- and third-order terms for DOI, the degree of internationalization.

Table 4 lists and defines the operationalization of all variables.

Variables

Dependent variables

Similarly to previous studies listed in Table 1, the financial performance measures chosen were:

- (1) ROS, return on global total sales: avoids the effects of different asset valuations resulting from the timing of investment or depreciation (Geringer *et al.*, 1989).
- (2) ROA, return on assets: has been widely used in many previous studies of the relationship between degree of internationalization and performance (Daniels and Bracker, 1989; Haar, 1989; Ramaswamy, 1995; Gomes and Ramaswamy, 1999).

Table 4 Descriptions of variables in this study

PERF	Financial performance variable (ROS, ROA)	
Firm size	Natural logarithm of total employees	
DOI	Index for degree of internationalization, the sum of eigenvector-weighted FSTS, FETE, and FOTO. Hence DOI has a theoretical range from 0 to 3. (This is a similar approach to several studies in the performance-multinationality literature listed in Table 1.)	
FSTS	Foreign sales/total sales	
FETE	Number of foreign employees/total number of employees	
FOTO	Number of foreign offices/total number of offices	
(DOI) ² and (DOI) ³	Quadratic and cubic terms, respectively	
I_m	Set of dummy variables to control subsector effects	
m	= 1, 2, ..., 5 for knowledge-based service sector = 1, 2, ..., 4 for capital-intensive service sector	
I	Represents the companies in the study (cross-sectional component)	
t	Corresponds to the different years (time-series component)	
ε_{it}	Error term	
Knowledge-based service subsectors (four subsectors)	Industry dummies	Using the publishing sector as the baseline variable, to reach lowest Cronbach's alphas
	I_1	1 = Advertising
	I_2	1 = Market research
	I_3	1 = Securities and financial service
Capital-based service subsectors (seven subsectors)	Industry dummies	Using the shipping and trucking sector as the baseline variable, to reach lowest Cronbach's alphas
	I_1	1 = Air transportation
	I_2	1 = Construction
	I_3	1 = Hotels
	I_4	1 = Restaurant and fast food chains
	I_5	1 = Retail trade
	I_6	1 = Wholesale trade
	Home country effect	Control for U.S.-based vs non-U.S.-based home nation
	C_1	1 = U.S.-based

Independent variables

The principal explanatory variable is the degree of internationalization (DOI), a three-component index for the degree of internationalization as suggested by Sullivan (1994a, b). DOI is the eigenvector-weighted sum of FSTS (foreign sales/total sales), FETE (number of foreign employees/number of total employees), and FOTO (number of foreign offices/number of total offices). A principal component analysis is applied to test whether all three dimensions are loaded on one single component, and then the component eigenvectors are treated as weights in deriving the combined DOI index (Gomes and Ramaswamy, 1999).

Control variables

Several studies have used additional *control variables* (such as research and development intensity, firm size, sector effect) to further investigate the relationship between multinationality and performance. These include the works of Buckley *et al.* (1977, 1984), Siddharthan and Lall (1982), Kumar (1984), Grant (1987), Daniels and Bracker (1989), Morck and Yeung (1991), Al-Obaidan and Scully (1995), Ramaswamy (1995), Tallman and Li (1996), Gomes and Ramaswamy (1999), and Kotabe *et al.* (2002).

Control variables used here, which are hypothesized to affect firm performance, are firm size, sector effect, and home country effect. Firm size, a common variable related to firm performance, was measured by the natural logarithm of total employees and used to control for economies and diseconomies of scale at the corporate level. Log transformation not only makes the results easy to interpret, because the changes in the logarithm domain represent relative (percentage) changes in the original metric; it also makes the distribution of data closer to normality.

Subsector effects have been confirmed by many prior researchers, and we segment the sample into knowledge-based and capital-intensive service sectors with dummy variables, as shown in Tables 2 and 3. To maximize the heterogeneity across the sector dummy variables, careful consideration has been given to the selection of sectors for regression analysis. The final testable sample includes companies from more than 10 different home countries.

Outliers and the range of DOI

DOI is an index for internationalization that has a theoretical range from 0 to 3. As it turned out, in the sample, almost all DOI values were at or below

1.0. The very few outliers (amounting to 1.95% of observations) with DOI > 1.0 were removed from the sample, as they could distort the results. The total number of observations, N , was only reduced from 618 to 606.

Results

Correlation data provided a first check on multicollinearity. Intercorrelations among the full set of predictor variables were sufficiently low to preclude the generation of unstable beta coefficients in regression analysis.

Initially, both ROS and ROA were tried in the regressions as dependent variables. However, the results from both are similar, and there is a moderate degree of collinearity between ROS and ROA. Hence it is worth showing only ROS as dependent variable. For all runs, a quadratic equation $[(DOI)^2]$ as well as a cubic equation $[(DOI)^2 \text{ with } (DOI)^3]$ was tried, the latter to test the sigmoid or S-shaped overall hypothesis.

Table 5 shows results for all 11 service sectors combined, even though there are *a priori* reasons (described above) for believing that the characteristics of knowledge-based and capital-intensive subsectors will be different. Table 6 shows results for just the capital-intensive subsectors, and Table 7 shows results for just the knowledge-based subsectors.

As multinationality can explain only a *part* of the financial performance of multinational companies, low adjusted R^2 s were expected. In the event, reasonably high adjusted R^2 s were realized, from 0.20 to 0.41 (see Tables 5–7). All models are strongly significant, with high F-values.

Comparisons of F-values and adjusted R^2 s indicate that results generally improved when the cubic variable $(DOI)^3$ was added to the quadratic specification. This confirms that the cubic term is not only significant in itself, but also adds to the overall results.

Results for the home-country dummy (a positive coefficient for C_1 in all significant values) confirm that service multinationals based in the U.S. are better able to reap the net positive benefits of internationalization than non-U.S. firms. This possibly reflects the fact that the expansion of the service sector was earlier, and more advanced, in the US than in most other nations.

Figures 2 and 3 show the fitted regression curves (quadratic and cubic) generated by the SPSS statistical program, corresponding to Tables 6 and 7, for the capital-intensive and knowledge-based sectors, respectively.

**Table 5** All service sectors (capital-intensive and knowledge-based)

ROS vs:	Model A1 (square)	Model A2 (cubic)
Intercept	0.0009	-0.0052
Firm size	0.122**	0.103**
DOI index	-0.426**	0.907***
(DOI index) ²	0.803***	-2.823***
(DOI index) ³		2.364***
<i>l</i> ₁	-0.160***	-0.210***
<i>l</i> ₂	-0.078	-0.111**
<i>l</i> ₃	0.282***	0.561***
<i>l</i> ₄	-0.166***	-0.174***
<i>l</i> ₅	0.025	0.026
<i>l</i> ₆	0.031	-0.052
<i>l</i> ₇	-0.394***	-0.395***
<i>l</i> ₈	-0.287***	-0.270***
<i>l</i> ₉	-0.066	-0.083*
<i>l</i> ₁₀	-0.280***	-0.293***
<i>C</i> ₁	0.126***	0.110***
Adjusted <i>R</i> ²	23.9%	27.4%
F-value	14.566***	16.196***
Number of observations, <i>N</i>	606	606

Firm size: logarithm of total employees; DOI index: index of degree of internationalization; (DOI index)²: Square item; (DOI index)³: Cubic item; *l*₁: sector effect (dummy, 1 = advertising, 0 = otherwise); *l*₂: sector effect (dummy, 1 = market research, 0 = otherwise); *l*₃: sector effect (dummy, 1 = securities and financial services, 0 = otherwise); *l*₄: sector effect (dummy, 1 = publishing, 0 = otherwise); *l*₅: sector effect (dummy, 1 = air transport, 0 = otherwise); *l*₆: sector effect (dummy, 1 = construction, 0 = otherwise); *l*₇: sector effect (dummy, 1 = hotel, 0 = otherwise); *l*₈: sector effect (dummy, 1 = restaurant and fast food chains, 0 = otherwise); *l*₉: sector effect (dummy, 1 = retail trade, 0 = otherwise); *l*₁₀: sector effect (dummy, 1 = wholesale, 0 = otherwise); *C*₁: home country effect (dummy, 1 = U.S.-based, 0 = otherwise). ****P* < 0.01; ***P* < 0.05; **P* < 0.1; all two-tailed tests. Eleven sectors are included (shipping and trucking serves as the baseline sector for dummy effect).

The quadratic (squared) models show a U-shaped relationship between performance and multinationality for the entire sample and for the capital-intensive subsample. An initial decline in performance (based on Stage 1 costs and barriers to initial internationalization) is followed by a subsequent improvement in performance. For knowledge-based sectors, the first-order and second-order terms in Model K₁ (Table 7) are positive, but the first-order coefficient is small. The quadratic plot in Figure 3 shows that, after an initially almost flat portion of the curve, ROS picks up positively for knowledge-based sectors.

The cubic models conform to the sigmoid hypothesis (showing *all three* stages of Figure 1) only for the knowledge-based subsectors (in Table 7). Indeed, for the capital-intensive sample and total

Table 6 Capital-intensive service sectors

ROS vs:	Model C ₁ (square)	Model C ₂ (cubic)
Intercept	-0.073	-0.160***
Firm size	0.180***	0.201***
DOI index	-0.739***	2.131***
(DOI index) ²	1.093***	-6.094***
(DOI index) ³		4.466***
<i>l</i> ₁	-0.220**	-0.174*
<i>l</i> ₂	-0.057	0.010
<i>l</i> ₃	0.112	0.120
<i>l</i> ₄	-0.085	0.011
<i>l</i> ₅	-0.063	0.024
<i>l</i> ₆	0.268**	0.232**
<i>C</i> ₁	0.098	0.104*
Adjusted <i>R</i> ²	20.2%	30.5%
F-value	8.843***	13.394***
Number of observations, <i>N</i>	364	364

Firm size: logarithm of total employees; DOI index: index of degree of internationalization; (DOI index)²: square item; (DOI index)³: cubic item; *l*₁: sector effect (dummy, 1 = air transport, 0 = otherwise); *l*₂: sector effect (dummy, 1 = construction, 0 = otherwise); *l*₃: sector effect (dummy, 1 = hotels, 0 = otherwise); *l*₄: sector effect (dummy, 1 = restaurant and fast food chains, 0 = otherwise); *l*₅: sector effect (dummy, 1 = retail trade, 0 = otherwise); *l*₆: sector effect (dummy, 1 = wholesale, 0 = otherwise); *C*₁: home country effect (dummy, 1 = U.S.-based, 0 = otherwise). ****P* < 0.01; ***P* < 0.05; **P* < 0.1; all two-tailed tests. Seven sectors are included (shipping and trucking serves as the baseline sector for dummy effect).

sample the signs of the coefficients for DOI are opposite to those expected. However, looking at Figure 2 for the capital-intensive subsectors (the total-sample plot is similar) we see that ROS is almost flat between DOI = 0 and DOI = 0.6, after which it increases quickly with a positive slope.

It is only in the knowledge-based subsectors (Figure 3) that, with DOI > 0.7, performance or ROS actually declines, confirming the expectations of the sigmoid theory's Stage 3 (as seen in Figure 1).

Discussion

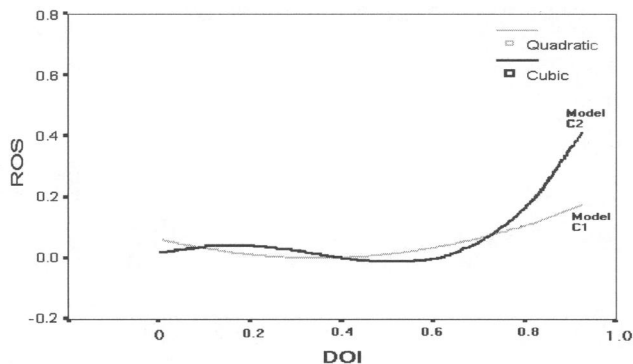
Why are knowledge-based service companies different from capital-intensive firms? First, comparing Figures 2 and 3, we find that knowledge-based sectors reap the positive (Stage 2) benefits of internationalization earlier (that is at a lower DOI). Second, knowledge-based sectors 'over-internationalize,' reaching the suboptimal Stage 3, whereas capital-intensive sectors do not.

In the update of their *internationalization* theory Johanson and Vahlne (1990) explicitly recognize that the business context is an important explanation for the different rates at which and extent to

Table 7 Knowledge-based service sectors

ROS vs:	Model K_1 (square)	Model K_2 (cubic)
Intercept	0.176***	0.168***
Firm size	-0.224**	-0.175**
DOI index	0.104	-0.774***
(DOI index) ²	0.359**	3.374***
(DOI index) ³		-2.226***
I_1	-0.522***	-0.436***
I_2	-0.463***	-0.392***
I_3	-0.111*	-0.116*
C_1	0.004	-0.004
Adjusted R^2	36.5%	41.2%
F-value	21.652***	23.042***
Number of observations, N	242	242

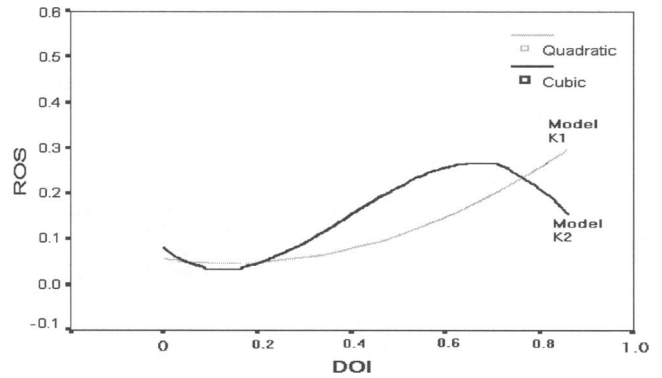
Firm size: logarithm of total employees; DOI index: index of degree of internationalization; (DOI index)²: square item; (DOI index)³: cubic item; I_1 : sector effect (dummy, 1 = advertising, 0 = otherwise); I_2 : sector effect (dummy, 1 = market research, 0 = otherwise); I_3 : sector effect (dummy, 1 = securities and financial services, 0 = otherwise); C_1 : home country effect (dummy, 1 = U.S.-based, 0 = otherwise). *** $P < 0.01$; ** $P < 0.05$; * $P < 0.1$; all two-tailed tests. Four sectors are included (publishing serves as the baseline sector in the regression for dummy effect).

**Figure 2** Fitted curve of ROS vs DOI: capital-intensive sectors.

which different firms internationalize. They, as well as Erramilli (1987), discuss existing network relationships and the advantages that some firms possess in being able to ‘follow the client’ abroad. Erramilli’s dissertation distinguished between *client-following* and *market-seeking* service multinationals. In knowledge-based service sectors, from the 1980s on, international expansion tended to be driven by a ‘follow the client’ strategy, as advertising, financial services and market research companies followed their multinational clients abroad.

Stages 1 and 2 of hypothesis

Thus knowledge-based services possess three advantages that enable them to reap the advantages

**Figure 3** Fitted curve of ROS vs DOI: knowledge-based sectors.

(ROS) of internationalization faster than capital-intensive sectors:

- (1) They have a lower burden of tangible asset investment. Petersen and Pedersen (1998) describe this in terms of a lower risk of committing ‘irreversible resources’ to each foreign market.
- (2) They have clients already established abroad.
- (3) They possibly have a greater global standardization, which lowers the cost per foreign entry, and enables the net benefits of foreign expansion to be reaped sooner.

We can see this by comparing Figures 2 and 3. Knowledge-based subsectors (Figure 3) show the net gains of internationalization (positive slope of ROS vs DOI) as early as DOI=0.17. By contrast, in Figure 2, the slope is not positive until DOI=0.55 (ignoring the very slight initial rise between DOI=0 and DOI=0.15). One can even draw the same conclusion by comparing the quadratic plots: knowledge-based subsectors reap the net positive benefits of international expansion sooner. (Possibly, this is also because the larger capital investments required in aircraft, construction, shipping, hotels, restaurants, etc. require a larger global scale before the net benefits of expansion are realized.)

Stage 3 of hypothesis

In comparing Figures 2 and 3, we find that only the knowledge-based subsectors show a downturn in performance (after approximately DOI>0.7). For capital-intensive subsectors the slope remains positive. The capital-intensive subsample (Figure 2) comprises firms that conform to Stages 1 and 2 of the hypothesis proposed in Figure 1 – but this subsample does *not* contain overly internationa-

lized Stage 3 multinationals that have expanded beyond their optimum point.

For capital-intensive service sectors (air transport, construction, hotels, restaurants, retail trade, wholesale, and shipping and trucking) the fixed asset risk and capital cost is much higher. Consequently, such sectors are likely to be more cautious in their international expansion and less likely to stray into Stage 3 (suboptimal zone for performance). This is only an *ex post* hypothesis.

By contrast, knowledge-based firms (advertising, market research, securities and diversified financial services, and publishing) can more easily 'over-expand' into the suboptimal Stage 3 (as seen in Figure 3 and in the negative sign cubic term in Table 7). Petersen *et al.* (2002), discussing similar new-economy firms, describe this as 'rash foreign market expansion' that leads to poor performance. To recapitulate the earlier discussion: it is unlikely that most firms would *deliberately* over-expand. Few companies possess the managerial tools (such as regression plotting of firms in a sector) that would tell them when they have over-internationalized. Hence, in a given cross-sectional analysis, some fraction of firms may be expected to have inadvertently strayed into the suboptimal Stage 3.

Clearly, our results confirm that it is only a small fraction of firms that are in Stage 3; there are none in the capital-intensive subsectors (Figure 2) and relatively few in the knowledge-based subsectors (Figure 3).

This confirms the subordinate hypothesis that most companies will be found in a zone (Stage 2) where there is a positive relationship between performance and multinationality – and there will be relatively few in Stages 1 and 3.

Conclusions and summary: a unified three-stage theory of international expansion

The bedrock assumption of international business studies, that greater multinationality augments the performance of firms (Stage 2), is reinforced in this study, but with two additional qualifications:

- (1) At an early stage (Stage 1) of internationalization there may be a diminution in performance because of the initial learning costs, cultural and foreign market inexperience, an insufficient scale of global operations, and, in general, the liability of foreignness.
- (2) Some firms, in some sectors, may 'over-internationalize' by expanding into too many

nations (Stage 3) and again suffer a net (or incremental) negative effect on performance, because their final expansion is into peripheral or small markets, and because coordination and governance costs increase faster than incremental revenues from further expansion.

Earlier literature showing U-shaped or inverted-U-shaped results had hinted at the existence of these three stages. This paper, for the first time, integrates these into a unified three-stage theory, as illustrated in Figure 1.

The empirical results broadly (although not completely) validate this idea. The knowledge-based service subsectors do conform to all three stages of the theory (as illustrated in Figure 3). Capital-intensive subsectors (as seen in Figure 2), however, exhibit only Stages 1 and 2. Moreover, for this subsector, significant net benefits of internationalization were not seen until much later in the internationalization process (compared with knowledge-based subsectors), possibly because aircraft, construction, shipping, hotels, restaurants, etc. require a larger global scale of operations before net benefits are realized.

By the same token, knowledge-based subsector firms, having less of a tangible capital assets burden, and moreover enjoying a 'follow the client' advantage, can reap the benefits of international expansion faster than capital-intensive subsectors can.

Stage 3 (the suboptimal zone resulting from over-internationalization), however, is relatively short in Figure 3, confirming the hypothesis that few companies stray into a suboptimal strategy.

The Hitt *et al.* (1994) hypothesis that the home country of the multinational can explain differences in performance, *ceteris paribus*, is confirmed in the positive sign for the U.S. dummy. Because the growth of the service sector was earlier and most extensive in the U.S. (compared with other nations), U.S.-based firms enjoyed a performance advantage in the time frame of the analysis.

Managerial implications

If managers were more explicitly conscious of the (temporarily) negative aspects of early internationalization (Stage 1) they could take ameliorative steps to lower the costs of initial international expansion. By the same token, in highly internationalized companies, if a firm was aware that it was reaching a limit to optimal performance growth, it might restrain further

expansion. But how is a company to know? One possibility is for the company to conduct econometric analyses like this one, on a cross-section of firms in their sector, to find the 'trend line' as in Figures 2 and 3, and compare their company's position against the regression mean. This kind of analysis can easily be performed in the econom-

ics or strategy departments of large multinationals. Companies lacking such analytical skills can outsource the analysis to consultants. Even if an econometric analysis were not done, at least the awareness of the dangers of over-expansion could prove to be a useful restraint in some firms.

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Appendix 1

Whereas the existing literature uses at least a curvilinear quadratic specification, we adopted a referee's strong suggestion to show, in Table 8, results using only a first-order specification for DOI. The three columns in the table, for Models A, C and K, correspond to Tables 5–7 above.

These results, as expected, show that performance is positively affected by multinationality. However, lacking the second- and third-order terms for DOI, the more complex relationship between these two variables cannot be seen.

Table 8 Results with linear specification for DOI only

ROS vs:	Model A ₀	Model C ₀	Model K ₀
Intercept	–0.00118	–0.142***	0.194***
Firm size	0.116**	0.206***	–0.278***
DOI index	0.424***	0.407***	0.442***
<i>l</i> ₁	–0.187***		–0.568***
<i>l</i> ₂	–0.101**		–0.528***
<i>l</i> ₃	0.188***		–0.126*
<i>l</i> ₄	–0.174***		
<i>l</i> ₅	0.025	–0.249**	
<i>l</i> ₆	–0.013	–0.144	
<i>l</i> ₇	–0.470***	0.065	
<i>l</i> ₈	–0.416***	–0.032	
<i>l</i> ₉	–0.152***	0.018	
<i>l</i> ₁₀	–0.328***	0.345***	
C ₁	0.138***	0.157**	0.003
Adjusted R ²	20.6%	15.4%	35.2%
F-value	13.195***	7.246***	20.827***
Number of observations, N	606	364	242

****P* < 0.01; ***P* < 0.05; **P* < 0.1; all two-tailed tests.